MASTER REPORT

Summit Care Centralized Identity & Hybrid Network Architecture

This document presents a mock enterprise architecture for Summit Care Medical Clinic. It shows how Azure, Microsoft Entra ID, on-premises Active Directory, one-way trusts, regional VPN hubs, Azure VPN Gateway, GPO logon restrictions, SSO, MFA, password hash sync, password writeback, RBAC, and HIPAA-aligned safeguards can support secure multi-campus healthcare operations without using any real company information.

SCGLOBAL.LOCAL | SUMMITCARE.ORG | MASTER-LEAF AD | HUB-AND-SPOKE VPN | HIPAA SAFEGUARDS

Project Goal

Centralize global identity in Azure while preserving each campus as an independent local Active Directory environment with its own users, groups, GPOs, resources, and operational boundaries.

Key Rule

Tier A global users can authenticate across approved campuses. Tier B local users stay restricted to their home campus through domain design and GPO logon controls.

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OVERVIEW

Purpose And Objectives

Summit Care Medical Clinic needs centralized identity management, secure cross-campus access for global staff, local autonomy for individual clinic domains, and a private WAN-like network built with VPNs. Azure hosts the central identity authority and provides the cloud security control plane.

- Centralize authentication and identity governance.
- Keep campus Active Directory environments independent.
- Allow Tier A global users to authenticate at approved campuses.
- Restrict Tier B local users to their assigned campus.
- Use regional hub-and-spoke VPN routing instead of direct campus-to-campus communication.
- Enable SSO, MFA, Conditional Access, password hash sync, password writeback, RBAC, and audit logging.

IDENTITY

Master-Leaf Active Directory Model

Layer	Summit Care Design	Purpose
Master domain	SCGLOBAL.LOCAL hosted in Azure	Central identity authority for Tier A global users and trust referrals.
Cloud tenant	summitcare.org in Microsoft Entra ID	Cloud identity, Microsoft 365 SSO, MFA, Conditional Access, password writeback, RBAC, and audit visibility.
East leaf domains	SCQNS.LOCAL, SCMAN.LOCAL, SCBKLYN.LOCAL, SCNWRK.LOCAL	Independent clinic domains for local users, GPOs, workstations, and local resources.
West leaf domains	SCPHX.LOCAL, SCDAL.LOCAL, SCDEN.LOCAL, SCLA.LOCAL	Independent clinic domains connected through the West regional hub.

Trust Model

Relationship	Behavior	Security Meaning
Campus to SCGLOBAL	Campus domains trust SCGLOBAL.LOCAL	Campus domain controllers can refer Tier A authentication requests to Azure.
SCGLOBAL to campus	SCGLOBAL does not trust campus domains	Local campus users cannot use SCGLOBAL as a bridge into other campuses.

Relationship	Behavior	Security Meaning
Campus to campus	No direct trust	Reduces lateral movement and preserves local campus isolation.

Identity Tiers

Tier	Stored In	Example	Access
Tier A - Global	SCGLOBAL.LOCAL	sdaniel@summitcare.org	Can authenticate at authorized campuses through one-way trusts.
Tier B - Local	Campus AD	frontdesk01@scqns.local	Restricted to the home campus only.
Privileged Admin	SCGLOBAL and Entra roles	cloud-admin@summitcare.org	Protected with MFA, RBAC, least privilege, and audit logs.

Tier A Authentication Flow

1. A Tier A user signs in as sdaniel@summitcare.org at a Dallas campus device.
2. SCDAL.LOCAL sees that the user is not a local Dallas account.
3. The Dallas domain controller uses the one-way trust to refer authentication to SCGLOBAL.LOCAL.
4. SCGLOBAL validates the credentials across the private VPN path.

5. The campus applies local authorization, workstation policy, and resource permissions.

NETWORK

Hub-and-Spoke VPN And IP Scheme

The network design uses a different private IP scheme than the reference material. Azure owns 10.90.0.0/16, the East region owns 10.91.0.0/16, and the West region owns 10.92.0.0/16.

Zone	CIDR	Example Systems
Azure identity backbone	10.90.0.0/16	SCGLOBAL DCs, sync server, management services, Azure VPN Gateway.
Azure identity subnet	10.90.10.0/24	SCGLOBAL-DC01 10.90.10.10, SCGLOBAL-DC02 10.90.10.11, SC-SYNC01 10.90.10.20.
Azure gateway subnet	10.90.254.0/27	Azure VPN Gateway for hub-to-cloud tunnels.
East region	10.91.0.0/16	Queens hub, Manhattan, Brooklyn, Newark.
West region	10.92.0.0/16	Phoenix hub, Dallas, Denver, Los Angeles.

Regional Routing

Region	Hub	Spokes	Path
East	Queens Hub 10.91.0.0/24	Manhattan 10.91.10.0/24, Brooklyn 10.91.20.0/24, Newark 10.91.30.0/24	Campus Meraki VPN to Queens, then Queens to Azure VPN Gateway.
West	Phoenix Hub 10.92.0.0/24	Dallas 10.92.10.0/24, Denver 10.92.20.0/24, Los Angeles 10.92.30.0/24	Campus Meraki VPN to Phoenix, then Phoenix to Azure VPN Gateway.

> Campuses do not communicate directly. The intended path is Campus to Meraki VPN, Meraki VPN to regional hub, regional hub to Azure VPN Gateway, then Azure to SCGLOBAL services.

Trust Network Requirements

- Every campus DNS server uses conditional forwarders for SCGLOBAL.LOCAL pointing to 10.90.10.10 and 10.90.10.11.
- VPN and firewall rules allow Kerberos, LDAP, DNS, and required domain controller communication between campus DCs and SCGLOBAL DCs.
- Domain controllers use reliable NTP because Kerberos authentication is time-sensitive.
- Campus domains create one-way outgoing trusts to SCGLOBAL.LOCAL.
- No campus creates a direct trust with another campus.

HYBRID SYNC

Microsoft Entra Hybrid Identity Sync Build

Prerequisite	Summit Care Standard	Validation
Active Directory health	AD 2016 or newer recommended, clean DNS, healthy replication.	Run dcdiag /v and repadmin /replsummary.
Cloud tenant	summitcare.org verified in Microsoft Entra ID.	Confirm DNS TXT verification in Entra admin center.
UPN alignment	Users sign in as user@summitcare.org where cloud identity is required.	No user@localdomain.local cloud sign-in names.
Sync server	SC-SYNC01, domain joined, static IP 10.90.10.20, patched, dedicated to sync.	A dedicated member server is preferred; a DC install can work in a lab but is not the production ideal.
Networking	Outbound HTTPS TCP 443 to Microsoft endpoints.	No SSL inspection or firewall policy breaks sync traffic.

Deployment Steps

1. Prepare Active Directory by fixing replication, DNS, duplicate user, duplicate mail, and duplicate proxyAddresses issues.
2. Verify summitcare.org in Microsoft Entra ID and plan licenses for Conditional Access, SSPR, and writeback features.
3. Configure the UPN suffix in Active Directory Domains and Trusts, then update users to match the verified cloud domain.
4. Build SC-SYNC01 as a domain-joined sync server with static IP, OS patches, and access to all required domain controllers.

5. Download Microsoft Entra Connect Sync from the Entra admin center.
6. Choose customized settings and connect with the required Entra and on-premises administrative permissions.
7. Select Password Hash Sync as the recommended authentication method. Enable Seamless SSO and password writeback where policy and licensing support them.
8. Select OUs to sync and filter out stale, disabled, service, or lab objects that should not appear in the cloud.
9. Run Start-ADSyncSyncCycle -PolicyType Initial and review Synchronization Service Manager.
10. Validate users, groups, UPNs, mail attributes, Microsoft 365 sign-in, Teams, Outlook, MFA, and password reset behavior.

Ongoing Commands And Checks

Task	Command Or Location	Purpose
AD health	dcdiag /v	Find domain controller, DNS, and service issues.
Replication summary	repadmin /replsummary	Confirm AD replication is healthy before sync.
Delta sync	Start-ADSyncSyncCycle -PolicyType Delta	Push normal recent changes.
Initial sync	Start-ADSyncSyncCycle -PolicyType Initial	Use after major scope or connector changes.
Sync review	miisclient.exe and Entra admin center	Review imports, exports, errors, and cloud sync state.

GPO ISOLATION

Cross-Domain Local Logon Restrictions

Summit Care uses GPO deny-local-logon controls to prevent local Tier B users from interactively logging into computers outside their assigned campus. This is separate from the Tier A global authentication model.

Example: Block Queens Users From Brooklyn Workstations

1. On the Brooklyn domain controller, open Active Directory Users and Computers.
2. Create a global security group named Deny_SCQNS_Login in SCBKLYN.LOCAL.
3. Add the Queens Domain Users group from SCQNS.LOCAL as a member.
4. Open Group Policy Management and create Deny SCQNS Logon To SCBKLYN.
5. Edit Computer Configuration, Policies, Windows Settings, Security Settings, Local Policies, User Rights Assignment.
6. Open Deny log on locally and add Deny_SCQNS_Login.
7. Link the GPO to the SCBKLYN_Workstations OU.
8. Run gpupdate /force on a Brooklyn workstation.
9. Test with a Queens Tier B account and confirm the sign-in method is not allowed.

Reverse Restriction

Step	Queens Domain Action	Purpose
Create group	Deny_SCBKLYN_Login	Defines Brooklyn users that should not log into Queens workstations.
Add members	Add SCBKLYN\Domain Users or a targeted Brooklyn user group.	Targets Brooklyn Tier B identities.
Create GPO	Deny SCBKLYN Logon To SCQNS	Documents the restriction in Group Policy.
Configure policy	Add Deny_SCBKLYN_Login to Deny log on locally.	Blocks interactive local sign-in.
Link and test	Link to SCQNS_Workstations and run gpupdate /force.	Confirms the restriction works before broad rollout.

> Do not deny Domain Admins, Enterprise Admins, local Administrators, SYSTEM, or emergency break-glass accounts. Apply these GPOs to workstation OUs, not domain controller OUs.

HIPAA

Security, Risk, And Governance

Safeguard Area	Azure / Entra Control	Summit Care Value
Unique user identification	UPN alignment, named accounts, Entra user inventory.	User actions map back to individual staff identities.
Person or entity authentication	MFA, PHS, Conditional Access, password policy.	Reduces risk from stolen passwords and weak sign-in behavior.
Access control	RBAC, groups, app assignment, Conditional Access.	Limits access to authorized staff and least-privilege admins.
Audit controls	Entra audit logs, sign-in logs, role assignment history.	Supports review of who accessed systems and what changed.
Access termination	Disable or delete workflow, session revocation, sync validation.	Improves offboarding speed and reduces lingering access.

Potential Blockers

Blocker	Risk	Mitigation
Regional hub failure	Spokes may lose their Azure path for Tier A authentication.	Use HA Meraki pairs, redundant ISPs, and failover testing.

Blocker	Risk	Mitigation
DNS resolution failure	Campus domains cannot locate SCGLOBAL.LOCAL domain controllers.	Use conditional forwarders to 10.90.10.10 and 10.90.10.11.
VPN Gateway bandwidth	Hub-to-cloud traffic can bottleneck.	Choose the correct Gateway SKU and monitor tunnel utilization.
Latency for remote regions	Authentication can feel slow across distant sites.	Choose a balanced Azure region or deploy another SCGLOBAL DC closer to the West hub.
Dirty AD before sync	Duplicates and replication issues cause sync failures.	Run dcdiag, repadmin, duplicate checks, and UPN cleanup first.

SUMMARY

Final Result

The final Summit Care architecture creates centralized cloud control without collapsing every clinic into one flat trust model. SCGLOBAL.LOCAL in Azure acts as the master identity authority, summitcare.org in Microsoft Entra ID provides cloud governance, regional VPN hubs provide private connectivity, and campus domains retain local autonomy. Tier A users can authenticate where authorized; Tier B users stay local. Azure and Entra add SSO, MFA, Conditional Access, password hash sync, password writeback, RBAC, audit logs, and HIPAA-aligned evidence. The design is secure, segmented, scalable, and ready to support many Summit Care Medical Clinic locations.